INCH-POUND

A-A-54879
16 July 1993

COMMERCIAL ITEM DESCRIPTION

BREATHING CIRCUIT, ANESTHESIA

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-R-37725B, which is cancelled.

This Commercial Item Description covers requirements for two types of anesthesia breathing circuits suitable for the administration of anesthesia.

Salient characteristics:

Scope. The breathing circuits covered by this document include the following types as specified:

Type I - Swivel connector
Type II - Rigid connector with adapter

Materials. The rebreathing bag shall be fabricated from latex rubber. The 22 millimeter (mm) bag inlet connector shall be fabricated from suitable flexible rubber or plastic. The breathing tubes shall be fabricated from flexible corrugated polyethylene or a combination of polyethylene and synthetic rubber. Filter assembly, swivel "Y" connector, rigid "Y" connector, right-angle elbow adapter, and connecting components and fittings shall be fabricated from materials suitable for the purpose intended.

Style and design. The breathing circuit shall consist of 0.5 micron bacteria-retentive filter, two flexible breathing tubes, a rebreathing bag of 3-liter capacity, and an integral "Y" connector component suitable for attachment to a face mask.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

The rebreathing bag shall be 13-1/2 inches long. The wall thickness of the bag shall be 0.010 inch minimum. The breathing tube shall be 34 to 40 inches long. The bacteriological filter shall be designed to prevent cross-contamination during administration of anesthesia. The filter inlet shall be capable of forming a secure, leakproof junction when connected to a standard 22 mm external tapered connection.

The two types of breathing circuits as specified shall be as follows:

Type I. The Type I unit shall possess a built in "Y" connector capable of 180 degrees minimum rotation. The exposed opening of the swivel connector shall be internally tapered to accept 15 mm endotracheal connectors or adapters that are in accordance with ASTM F1243 in a leakproof manner. The breathing tubes shall be securely attached to the upper arms of the swivel "Y" connector.

Type II. The Type II unit shall possess a built-in rigid "Y" connector securely attached to both breathing tubes, and a right-angle elbow adapter assembled to the patient end of the "Y" (lower leg). The elbow adapter shall form a completely rotatable mask connection. The opening of the rigid "Y" connector shall be internally tapered to accept a 15 mm endotracheal connectors or adapters that are in accordance with ASTM F1243, in a leakproof manner. The patient end of the right-angle adapter shall possess a standard tapered 22 mm external connection and a 15 mm internal tapered connection to accommodate 15 mm endotracheal connectors or adapters that are in accordance with ASTM F1243. The machine end of the adapter shall form a standard external 15 mm connection, tapered in accordance with ASTM F1243, and form a tight, leakproof junction with the rigid "Y" connector.

Identification marking. The name or registered trademark of the manufacturer shall appear on at least one of the breathing circuits components.

Performance. The opening of the swivel or rigid "Y" connector and the external 15 mm end of the right-angle adapter shall pass the gauging test specified in ASTM F1243.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:39Z
Check the source to verify that this is the currentversion before use.

The rebreathing bag, when subjected to an internal air pressure of 25 mm Hg, shall show no evidence of leaking or rupture. Further, when the bag is placed into an air oven for seven days at a temperature of 156° to 160°F (68.9° to 71.1°C), it shall not become tacky or brittle or show decrease in tensile strength by more than 25 percent.

When the breathing circuit is fully assembled, a 10 lb. pull at any junction will not separate or loosen any component. When tested as follows, leakage shall not exceed 100 cubic centimeter per minute. A suitably calibrated flowmeter and mercury manometer shall be used to test leakage of the breathing circuit proper. The filter assembly shall be assembled to one of the tubes, if not permanently attached to the rebreathing tube, and the right-angle elbow adapter attached to the "Y" connector (when applicable). With the opening of the filter assembly and the swivel "Y" connector (Type I unit) or the right-angle elbow adapter (Type II unit) each suitably plugged air pressure of 15 mm Hg shall be internally admitted to the opening of the other breathing tube. The manometer shall be inserted into the circuit between the controlled air supply and the breathing tube opening, with the calibrated flowmeter between the breathing tube opening and the manometer. The stabilized flow, as indicated by the flowmeter, shall be recorded.

Workmanship. The breathing circuit shall be free from holes, cracks, tears, sharp edges, blocked openings, foreign matters and any other defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing twenty individually package breathing circuit of the type specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:39Z
Check the source to verify that this is the current version before use.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one breathing circuit as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one breathing circuit. Inspection level shall be S-1, and acceptance numbers shall be zero.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers, Item Identifications, and Types are covered by this document:

NATIONAL STOCK NUMBER	TYPE	ITEM IDENTIFICATION
6515-01-125-3239	I	BREATHING CIRCUIT, GAS ANESTHESIA, SWIVEL "Y", 20s

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:39Z
Check the source to verify that this is the current version before use.

A-A-54879

NATIONAL STOCK NUMBER	TYPE	ITEM IDENTIFICATION
6515-01-125-6592	II	BREATHING CIRCUIT, GAS ANESTHESIA, RIGID "Y", 20s

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5510
Location: c:\word\doc\gandhi\breathcd

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54879	2. DOCUMENT DATE (YYMMDD): 930716

3. DOCUMENT TITLE: BREATHING CIRCUIT, ANESTHESIA

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial / (2) AUTOVON
Defense Personnel Support Center ATTN: DPSC-MSE	(215)7372870 / 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — *Previous editions are obsolete* — 198-290

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:39Z
Check the source to verify that this is the current version before use.